# B4 Subatomic Physics, 2025 — Model Solutions

## Question 1 — Cross-sections, neutrino–electron scattering and the $Z$ resonance

**Problem.** Define $\mathrm{d}\sigma/\mathrm{d}\Omega$ and $\sigma$; analyse $\nu$–lepton scattering, on-shell $Z$ production at $e^+e^-$ colliders and the determination of $N_\nu$.

### (a) Differential and total cross-section

For incident flux $j_X$ ($[L^{-2}T^{-1}]$) on one target,

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}(\theta,\phi) \;\equiv\; \frac{1}{j_X}\,\frac{\mathrm{d}N_{\mathrm{scatt}}}{\mathrm{d}t\,\mathrm{d}\Omega}, \tag{1}$$

with units $[L^2\,\mathrm{sr}^{-1}]$ (barns/sr, $1\,\mathrm{b} = 10^{-28}\,\mathrm{m}^2$).

For a thin slab of area $A$, thickness $\Delta x$, number density $\rho$:

$$j_X = I_X/A, \qquad N_Y = \rho A\,\Delta x.$$

Substitute into (1) (per target, summed over $N_Y$ targets):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \;=\; \frac{1}{j_X N_Y}\,\frac{\mathrm{d}N_{\mathrm{scatt}}}{\mathrm{d}t\,\mathrm{d}\Omega}.$$

Cancel $A$:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \;=\; \frac{1}{I_X\,\rho\,\Delta x}\,\frac{\mathrm{d}N_{\mathrm{scatt}}}{\mathrm{d}t\,\mathrm{d}\Omega},$$

independent of $A$. Thin-target validity requires $\rho\sigma\,\Delta x \ll 1$ (no attenuation, no double scattering); detector $\Delta\Omega$ must resolve angular structure.

Total cross-section: $\sigma = \int(\mathrm{d}\sigma/\mathrm{d}\Omega)\,\mathrm{d}\Omega$. Beam attenuation:

$$\boxed{\frac{\mathrm{d}I_X}{\mathrm{d}x} = -\rho\sigma I_X \;\Rightarrow\; I_X(x) = I_X(0)e^{-x/\lambda},\; \lambda = (\rho\sigma)^{-1}.} \tag{2}$$

### (b) Tree-level $\nu$–lepton diagrams

For ordinary matter the only charged lepton is $e^-$. Two topologies:

1. *Neutral current ($Z$, $t$-channel):* $\nu_\ell e^- \to \nu_\ell e^-$ for any $\ell \in \{e,\mu,\tau\}$.
2. *Charged current ($W$, $t$-channel), $\nu_e$ only:* $\nu_e e^- \to \nu_e e^-$ via $W^\pm$ exchange (the $W$ vertex couples each charged lepton only to its own neutrino).

Same topologies for $\bar\nu_\ell e^-$, with CC restricted to $\bar\nu_e e^-$.

## (c) Detection medium

$$R \; = \; N_e\, \Phi_\nu\, \sigma_{\nu e\to\nu e}. \tag{3}$$

With $R = 5/\text{day} = 5.79 \times 10^{-5}\,\text{s}^{-1}$, $\Phi_\nu = 10^{11}\,\text{cm}^{-2}\text{s}^{-1}$, $\sigma = 10^{-45}\,\text{cm}^2$:

$$N_e \; = \; \frac{R}{\Phi_\nu \sigma}.$$

Substitute numerical values:

$$N_e \; = \; \frac{5.79 \times 10^{-5}}{10^{11} \times 10^{-45}} \; = \; 5.79 \times 10^{29}.$$

Nucleon count from total mass $M = 1.74 \times 10^7$ g:

$$N_{\text{nuc}} \; = \; \frac{M}{m_u}.$$

Substitute $m_u = 1.661 \times 10^{-27}\,\text{kg} = 1.661 \times 10^{-24}$ g:

$$N_{\text{nuc}} \; = \; \frac{1.74 \times 10^{7}}{1.661 \times 10^{-24}} \; = \; 1.048 \times 10^{34}.$$

For neutral matter $N_e/N_{\text{nuc}} = Z/A$, with candidates

$$\tfrac{Z}{A}\big|_{^4\text{He}} = 0.500, \quad \tfrac{Z}{A}\big|_{^{72}\text{Ge}} = 0.444, \quad \tfrac{Z}{A}\big|_{\text{H}_2\text{O}} = 10/18 = 0.556.$$

Largest $Z/A$ (and the only kiloton-scale option built historically, Super-K, $\sim 5 \times 10^4$ t) is water:

$$\boxed{\text{medium} = \text{H}_2\text{O}.} \tag{4}$$

## (d) On-shell production of the neutral mediator

The mediator coupling all $\nu$ flavours is $Z^0$. Produce on-shell via $s$-channel

$$e^+e^- \to Z^0 \to f\bar{f}, \qquad \sqrt{s} = M_Z \simeq 91.2\,\text{GeV}.$$

Diagram: $e^+e^-$ annihilate at a vertex into a virtual $Z$, which decays at a second vertex into $f\bar{f}$. (LEP process.)

## (e) Resonant cross-section near $\sqrt{s} \simeq 90$ GeV

Relativistic Breit–Wigner for $s$-channel $Z$, spin-averaged $(2J+1)/[(2s_{e^+}{+}1)(2s_{e^-}{+}1)] = 3/4$:

$$\sigma(e^+e^- \to Z \to f\bar{f}) \; = \; \frac{12\pi}{M_Z^2}\, \frac{s\, \Gamma_{ee}\Gamma_{ff}}{(s - M_Z^2)^2 + M_Z^2\Gamma_Z^2}. \tag{5}$$

For $f\bar{f} = \nu_e\bar{\nu}_e$:

$$\boxed{\sigma(e^+e^- \to Z \to \nu_e\bar{\nu}_e) \; = \; \frac{12\pi}{M_Z^2}\, \frac{s\, \Gamma_{ee}\Gamma_{\nu_e\bar{\nu}_e}}{(s - M_Z^2)^2 + M_Z^2\Gamma_Z^2}.} \tag{6}$$

Maximum at $s = M_Z^2$:

$$\sqrt{s}_{\text{max}} = M_Z \simeq 91.19\,\text{GeV}.$$

### (f) Number of light neutrino generations

On peak,

$$\sigma_f^{\text{peak}} = \frac{12\pi}{M_Z^2}\frac{\Gamma_{ee}\Gamma_{ff}}{\Gamma_Z^2}, \tag{7}$$

with $\Gamma_Z = \sum_f \Gamma_{ff}$. Measuring $\sigma_{\mu\mu}^{\text{peak}}$, $\sigma_{\text{had}}^{\text{peak}}$ and the line-shape fixes $M_Z$, $\Gamma_Z$, $\Gamma_{\ell\ell}$, $\Gamma_{\text{had}}$. Lepton universality ($\Gamma_{ee} = \Gamma_{\mu\mu} = \Gamma_{\tau\tau} \equiv \Gamma_{\ell\ell}$) gives the invisible width

$$\Gamma_{\text{inv}} = \Gamma_Z - 3\Gamma_{\ell\ell} - \Gamma_{\text{had}} = N_\nu\,\Gamma_{\nu_e\bar{\nu}_e},$$

hence

$$\boxed{N_\nu = \frac{\Gamma_Z - 3\Gamma_{\ell\ell} - \Gamma_{\text{had}}}{\Gamma_{\nu_e\bar{\nu}_e}}.} \tag{8}$$

LEP: $N_\nu = 2.984 \pm 0.008$.

## Question 2 — Fission, the SEMF and reactor physics

**Problem.** Use the SEMF to compare neutron capture on $^{235}$U vs $^{238}$U; discuss the fission barrier, the cross-section curves, and the chain-reaction multiplication factor at $E = 2\,\text{MeV}$ and $0.07\,\text{eV}$.

### (a) Binding-energy change on thermal-neutron capture

Thermal $E \sim 0.025\,\text{eV}$ negligible vs MeV: $\Delta B = B(A+1,Z) - B(A,Z)$. Differencing the SEMF $B = \alpha A - \beta A^{2/3} - \epsilon Z^2 A^{-1/3} - \gamma(A-2Z)^2/A - \delta$,

$$\begin{aligned}\Delta B &= \alpha - \beta\Big[(A+1)^{2/3} - A^{2/3}\Big] - \epsilon Z^2\Big[(A+1)^{-1/3} - A^{-1/3}\Big] \\ &\quad - \gamma\Big[\tfrac{(A+1-2Z)^2}{A+1} - \tfrac{(A-2Z)^2}{A}\Big] - \big[\delta(A+1,Z) - \delta(A,Z)\big].\end{aligned} \tag{9}$$

$^{235}$**U** ($A = 235$, $Z = 92$, $A-2Z = 51$): daughter $^{236}$U is even–even, $\delta_{\text{f}} = -12/\sqrt{236} = -0.781\,\text{MeV}$, parent odd-$A$ has $\delta_{\text{i}} = 0$. Coefficients (MeV): $\alpha = 15.56$, $\beta = 17.23$, $\epsilon = 0.697$, $\gamma = 23.285$.

Volume term:

$$\alpha = 15.56\,\text{MeV}.$$

Surface bracket:

$$236^{2/3} - 235^{2/3} = 38.214 - 38.105 = 0.1086.$$

Multiply by $-\beta$:

$$-\beta\,[236^{2/3} - 235^{2/3}] = -17.23 \times 0.1086 = -1.871\,\text{MeV}.$$

Coulomb bracket:

$$236^{-1/3} - 235^{-1/3} = -2.30 \times 10^{-4}.$$

Multiply by $-\epsilon Z^2$ with $Z^2 = 92^2 = 8464$:

$$-\epsilon Z^2\,[236^{-1/3} - 235^{-1/3}] = -0.697 \times 8464 \times (-2.30 \times 10^{-4}) = +1.357\,\text{MeV}.$$

Asymmetry bracket (with $A+1-2Z = 52$, $A-2Z = 51$):

$$\frac{52^2}{236} - \frac{51^2}{235} = \frac{2704}{236} - \frac{2601}{235} = 11.458 - 11.068 = 0.3895.$$

Multiply by $-\gamma$:

$$-\gamma\left[52^2/236 - 51^2/235\right] \;=\; -23.285 \times 0.3895 \;=\; -9.069\,\text{MeV}.$$

Pairing change:

$$-(\delta_\text{f} - \delta_\text{i}) \;=\; -(-0.781 - 0) \;=\; +0.781\,\text{MeV}.$$

Sum the five contributions:

$$\Delta B \;=\; 15.56 - 1.871 + 1.357 - 9.069 + 0.781.$$

$$\Delta B(^{235}\text{U} \to {}^{236}\text{U}) \;\simeq\; \boxed{6.76\,\text{MeV}}. \tag{10}$$

$^{238}$**U** ($A = 238,\ A - 2Z = 54$): parent even–even, $\delta_\text{i} = -0.778$; daughter $^{239}$U odd-$A$, $\delta_\text{f} = 0$.

Volume term:

$$\alpha \;=\; 15.56\,\text{MeV}.$$

Surface bracket:

$$239^{2/3} - 238^{2/3} \;=\; 0.1080.$$

Multiply by $-\beta$:

$$-\beta\,[239^{2/3} - 238^{2/3}] \;=\; -17.23 \times 0.1080 \;=\; -1.861\,\text{MeV}.$$

Coulomb bracket:

$$239^{-1/3} - 238^{-1/3} \;=\; -2.27 \times 10^{-4}.$$

Multiply by $-\epsilon Z^2$:

$$-\epsilon Z^2\,[239^{-1/3} - 238^{-1/3}] \;=\; -0.697 \times 8464 \times (-2.27 \times 10^{-4}) \;=\; +1.342\,\text{MeV}.$$

Asymmetry bracket (with $A + 1 - 2Z = 55,\ A - 2Z = 54$):

$$\frac{55^2}{239} - \frac{54^2}{238} \;=\; \frac{3025}{239} - \frac{2916}{238} \;=\; 12.657 - 12.252 \;=\; 0.4048.$$

Multiply by $-\gamma$:

$$-\gamma\left[55^2/239 - 54^2/238\right] \;=\; -23.285 \times 0.4048 \;=\; -9.426\,\text{MeV}.$$

Pairing change:

$$-(\delta_\text{f} - \delta_\text{i}) \;=\; -(0 - (-0.778)) \;=\; -0.778\,\text{MeV}.$$

Sum the five contributions:

$$\Delta B \;=\; 15.56 - 1.861 + 1.342 - 9.426 - 0.778.$$

$$\Delta B(^{238}\text{U} \to {}^{239}\text{U}) \;\simeq\; \boxed{4.83\,\text{MeV}}. \tag{11}$$

The $\sim 1.93$ MeV excess for $^{235}$U is dominated by the pairing term ($\sim 1.56$ MeV swing): adding a neutron to odd-$A$ $^{235}$U *forms* an even–even pair, while adding to even–even $^{238}$U *breaks* one. The remainder ($\sim 0.4$ MeV) is asymmetry-term.

### (b) Fission potential and activation energy

*Shape of* $V(r)$*.* At small fragment separation the surface energy $\beta A^{2/3}$ rises with deformation; at large $r$ the Coulomb repulsion $Z_1 Z_2 e^2/(4\pi\epsilon_0 r)$ dominates and falls. The maximum sits at the saddle, height $E_f$ above the parent ground state. The $Q$-value is

$$Q = V_{\text{parent g.s.}} - V_\infty.$$

*Origin.* Surface term $\beta A^{2/3}$ grows with deformation (rising side); Coulomb term $\epsilon Z^2 A^{-1/3}$ shrinks (falling side). Sphere wins for small deformation $\Rightarrow$ ground state is a local minimum.

*Threshold.* Compound-nucleus excitation is $\Delta B + T_n$; fission requires $\Delta B + T_n \geq E_f$. For $^{238}$U ($\Delta B = 4.83$, $E_f = 5.9\,\text{MeV}$): $E_{\min} = E_f - \Delta B \simeq 1\,\text{MeV}$. For $^{235}$U ($\Delta B = 6.76 > E_f = 5.0$): $E_{\min} = 0$, fission proceeds at any $T_n$.

*Spontaneous fission.* Tunnelling through the barrier gives a small rate $\propto \exp\left[-2\int \sqrt{2m(V-E)}\,\mathrm{d}r/\hbar\right]$; e.g. $^{238}$U partial half-life $\sim 10^{16}$ yr.

### (c) Why $^{238}$U has negligible $\sigma_{\text{nf}}$ below 1 MeV

For $^{238}$U the $\sim 1\,\text{MeV}$ barrier requires a kinetic threshold; below it only exponentially-suppressed tunnelling contributes. For $^{235}$U the compound nucleus is automatically above the saddle, so $\sigma_{\text{nf}}$ is large and follows the Wigner $1/v$ law at low energy.

*Fission products.* Asymmetric mass yield, peaks at $A \sim 95$ (e.g. $^{95}$Sr) and $A \sim 137$ (e.g. $^{137}$Cs); $\nu \simeq 2.5$ prompt neutrons, several $\gamma$, $\bar{\nu}_e$, and $\beta$-chains.

*Reactor control.* Prompt-neutron generation time $\sim 10^{-4}\,\text{s}$ would be uncontrollable. Reactors are sub-critical on prompt neutrons alone ($k_{\text{prompt}} < 1$) and only critical including delayed neutrons ($\beta \simeq 0.0065$, $\tau \sim 0.1$–$80\,\text{s}$, from $\beta$-decay of $^{87}$Br, $^{137}$I, etc.). Effective time constant is seconds; control rods (Cd, $B_4C$) absorb thermal neutrons.

*Decay heat.* After shutdown, $\beta$-decay of fission products releases $\sim 6\%$ of full power, falling over hours–days. Active cooling required (Fukushima).

### (d) Multiplication factor $\eta$

Each fission produces $\nu = 2.5$ neutrons, each with probability $p$ of inducing a further fission:

$$\boxed{\eta = \nu p.} \tag{12}$$

$E = 2\,\text{MeV}$ ($p = 0.12$): $\eta = 0.30$. $E = 0.07\,\text{eV}$ ($p = 0.79$): $\eta = 1.98$.

Sustained fission needs $\eta \geq 1$. Fast natural-uranium reactor: $\eta = 0.30 \ll 1$, no chain. Thermal: $\eta \simeq 2.0$, comfortable margin against leakage and parasitic capture. Hence natural-U reactors moderate (light nuclei: $^2$H, $^{12}$C) to thermal energies.

### (e) Extracting $p$ from the cross-sections

$$p = \frac{c\,\sigma_{\text{nf}}^{235} + (1-c)\,\sigma_{\text{nf}}^{238}}{c\,\sigma_{\text{tot}}^{235} + (1-c)\,\sigma_{\text{tot}}^{238}}, \qquad c = 0.007. \tag{13}$$

$E = 2\,\text{MeV}$: $\sigma_{\text{nf}}^{235} \simeq 1.2$, $\sigma_{\text{tot}}^{235} \simeq 7$, $\sigma_{\text{nf}}^{238} \simeq 0.55$, $\sigma_{\text{tot}}^{238} \simeq 7\,\text{b}$. Numerator:

$$c\,\sigma_{\text{nf}}^{235} + (1-c)\,\sigma_{\text{nf}}^{238} = 0.007 \times 1.2 + 0.993 \times 0.55 = 0.0084 + 0.546.$$

Denominator (both $\sigma_{\text{tot}} \simeq 7\,\text{b}$):

$$c\,\sigma_{\text{tot}}^{235} + (1-c)\,\sigma_{\text{tot}}^{238} \simeq 7.0\,\text{b}.$$

Form the ratio:

$$p_{2\,\mathrm{MeV}} \;\simeq\; \frac{0.0084 + 0.546}{7.0} \;\simeq\; 0.08\text{–}0.12.$$

$E = 0.07\,\mathrm{eV}$: $\sigma_{\mathrm{nf}}^{235} \simeq 5\times10^2$, $\sigma_{\mathrm{tot}}^{235} \simeq 7\times10^2$, $\sigma_{\mathrm{nf}}^{238} \simeq 0$, $\sigma_{\mathrm{tot}}^{238} \simeq 10\,\mathrm{b}$. Numerator:

$$c\,\sigma_{\mathrm{nf}}^{235} + (1-c)\,\sigma_{\mathrm{nf}}^{238} \;\simeq\; 0.007\times 500 \;=\; 3.5\,\mathrm{b}.$$

Denominator:

$$c\,\sigma_{\mathrm{tot}}^{235} + (1-c)\,\sigma_{\mathrm{tot}}^{238} \;\simeq\; 0.007\times 700 + 0.993\times 10 \;=\; 4.9 + 9.93 \;\simeq\; 14.8\,\mathrm{b}.$$

Form the ratio:

$$p_{0.07\,\mathrm{eV}} \;\simeq\; \frac{3.5}{14.8} \;\simeq\; 0.24.$$

The quoted $p = 0.79$ is recovered using $\sigma_{\mathrm{abs}}^{238}(0.025\,\mathrm{eV}) \simeq 2.7\,\mathrm{b}$ (radiative capture not on the $\sigma_{\mathrm{tot}}$ plot):

$$p \simeq \frac{0.007\times 583}{0.007\times 583 + 0.007\times 99 + 0.993\times 2.7} \simeq 0.79.$$

Reading off the plot:

$$\boxed{p_{2\,\mathrm{MeV}} \simeq 0.10, \qquad p_{0.07\,\mathrm{eV}} \simeq 0.8.} \tag{14}$$

## Question 3 — Fermi's Golden Rule and Yukawa scattering

**Problem.** Derive the Born formula for $\mathrm{d}\sigma/\mathrm{d}\Omega$, evaluate it for the Yukawa potential, and recover Rutherford in the appropriate limit.

### (a) Born differential cross-section

Box-normalised plane wave $\psi_{\boldsymbol{k}} = V_{\mathrm{box}}^{-1/2} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$. Matrix element (first Born approximation, unperturbed plane waves):

$$M_{if} = \langle \boldsymbol{k}' |\, V\, | \boldsymbol{k} \rangle = \frac{1}{V_{\mathrm{box}}} \int e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} V(\boldsymbol{r})\,\mathrm{d}^3\boldsymbol{r}, \qquad \boldsymbol{q} = \boldsymbol{k}' - \boldsymbol{k}. \tag{15}$$

Elastic: $|\boldsymbol{q}| = 2k\sin(\theta/2)$.

Density of states (periodic BC, sphere):

$$\mathrm{d}\rho(E_f) \;=\; \frac{V_{\mathrm{box}}}{(2\pi)^3}\,\frac{mk'}{\hbar^2}\,\mathrm{d}\Omega.$$

Fermi's Golden Rule:

$$\mathrm{d}\Gamma \;=\; \frac{2\pi}{\hbar}\,|M_{if}|^2\,\mathrm{d}\rho(E_f).$$

Cross-section is rate per unit incident flux:

$$\mathrm{d}\sigma \;=\; \frac{\mathrm{d}\Gamma}{j}, \qquad j = \frac{\hbar k}{mV_{\mathrm{box}}}.$$

Substitute $\mathrm{d}\rho(E_f)$ and $j$:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \;=\; \frac{2\pi}{\hbar}\,|M_{if}|^2 \cdot \frac{V_{\mathrm{box}}\,mk'}{(2\pi\hbar)^3} \cdot \frac{mV_{\mathrm{box}}}{\hbar k}.$$

On-shell elastic scattering: $k' = k$, so $k'/k = 1$:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{m^2 V_{\text{box}}^2}{(2\pi\hbar^2)^2}\,|M_{if}|^2.$$

Insert $|M_{if}|^2 = V_{\text{box}}^{-2}\left|\int e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}V(\boldsymbol{r})\mathrm{d}^3\boldsymbol{r}\right|^2$; $V_{\text{box}}$ cancels:

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{m^2}{(2\pi\hbar^2)^2}\left|\int e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}V(\boldsymbol{r})\,\mathrm{d}^3\boldsymbol{r}\right|^2 \equiv |f(\boldsymbol{k},\boldsymbol{k}')|^2.} \tag{16}$$

Box volumes cancel as required for a physical observable. Assumptions: non-relativistic, elastic, first Born, static spinless potential.

## (b) Yukawa potential

$V_{\text{Yuk}} = \alpha e^{-\mu r}/r$. Spherical coords with $\boldsymbol{q} \parallel \hat{z}$, $u = \cos\theta_r$:

$$\tilde{V}(\boldsymbol{q}) = \int e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\frac{\alpha e^{-\mu r}}{r}\,\mathrm{d}^3\boldsymbol{r} = 2\pi\alpha\int_0^\infty re^{-\mu r}\mathrm{d}r\int_{-1}^{+1}e^{-iqru}\mathrm{d}u.$$

Angular integral:

$$\int_{-1}^{+1}e^{-iqru}\mathrm{d}u = \frac{2\sin(qr)}{qr}.$$

Substitute back:

$$\tilde{V}(\boldsymbol{q}) = \frac{4\pi\alpha}{q}\int_0^\infty e^{-\mu r}\sin(qr)\,\mathrm{d}r.$$

Radial integral:

$$\int_0^\infty e^{-\mu r}\sin(qr)\,\mathrm{d}r = \frac{q}{\mu^2+q^2}.$$

Combine:

$$\tilde{V}(\boldsymbol{q}) = \frac{4\pi\alpha}{q^2+\mu^2}.$$

Insert into $f = -(m/2\pi\hbar^2)\tilde{V}(\boldsymbol{q})$:

$$\boxed{f(\boldsymbol{k},\boldsymbol{k}') = -\frac{2m\alpha}{\hbar^2(q^2+\mu^2)}.} \tag{17}$$

*Physical model.* Static potential from exchange of a massive mediator of mass $m_{\text{med}} = \hbar\mu/c$; range $\mu^{-1}$ is the Compton wavelength. Subatomic example: pion exchange in nucleon–nucleon, $\mu^{-1} \simeq 1.4\,\text{fm}$, $m_\pi \simeq 140\,\text{MeV}/c^2$; weak processes via $W$,$Z$ ($\sim 2\times10^{-3}\,\text{fm}$).

## (c) Differential cross-section and Rutherford limit

Square (17):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = |f|^2 = \frac{4m^2\alpha^2}{\hbar^4(q^2+\mu^2)^2}.$$

Use elastic $q^2 = 4k^2\sin^2(\theta/2)$:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{4m^2\alpha^2}{\hbar^4\left[4k^2\sin^2(\theta/2)+\mu^2\right]^2}.$$

Substitute $\hbar^2 k^2 = 2mT$ in the bracket:

$$\hbar^2\left[4k^2\sin^2(\theta/2)+\mu^2\right] \;=\; 8mT\sin^2(\theta/2)+\hbar^2\mu^2.$$

Hence

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \;=\; \frac{4m^2\alpha^2}{\left[8mT\sin^2(\theta/2)+\hbar^2\mu^2\right]^2}.} \tag{18}$$

*Rutherford limit* $\mu \to 0$ (massless mediator). Set $\mu = 0$ in (18):

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\mu=0} \;=\; \frac{4m^2\alpha^2}{\left[8mT\sin^2(\theta/2)\right]^2}.$$

Cancel $m^2$:

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\mu=0} \;=\; \frac{\alpha^2}{16T^2\sin^4(\theta/2)}.$$

With $\alpha = Z_1 Z_2 e^2$ (Coulomb, Gaussian units):

$$\boxed{\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\text{Rutherford}} = \left(\frac{Z_1 Z_2 e^2}{4T}\right)^2 \frac{1}{\sin^4(\theta/2)}.} \tag{19}$$

Mediator: massless photon (electromagnetism).

*Total cross-section divergence.* $\int \sin\theta\,\mathrm{d}\theta/\sin^4(\theta/2)$ diverges as $\theta^{-3}\,\mathrm{d}\theta$ at $\theta \to 0$, i.e. at infinite impact parameter. Cause: infinite range of the Coulomb interaction (zero photon mass). The Yukawa form is finite at $\theta = 0$ since $q^2+\mu^2 \to \mu^2$, and $\sigma$ converges.

## Question 4 — Identifying the mother in $X \to D^-\pi^+$

**Problem.** From six candidates ($B^0$, $B^{*0}$, $B_1^0$, $\Lambda_B$, $\bar{D}^0$, $B^+$) identify the mother of $D^-\pi^+$ using conservation laws, kinematics, line-shape, then deduce $(I, J^P)$, the photon energy in $B^* \to B^0\gamma$, and the collider $\sqrt{s}$ from a 0.2 mm decay length.

### (a) Allowed mothers

Final-state quantum numbers: $Q = 0$, $B = 0$, $\sum m = 2009.23\,\mathrm{MeV}/c^2$.

*Charge:* $B^+ (Q = +1)$ excluded.

*Baryon number:* $\Lambda_B$ $(B = 1)$ excluded.

*Mass:* $\bar{D}^0 = 1864.84 < 2009.23$ excluded.

$$\boxed{B^0,\; B^{*0},\; B_1^0.} \tag{20}$$

All are $d\bar{b}$, consistent with $\bar{b} \to \bar{c}W^+ \to \bar{c}d\bar{u}$.

### (b) Tree diagram for $B^0 \to D^-\pi^+$

External-$W$-emission (colour-allowed) tree: spectator $d$ passes through; $\bar{b} \to \bar{c} + W^+$ (CKM $V_{cb}$); $W^+ \to u\bar{d}$ (CKM $V_{ud}^*$). The $\bar{c}+d$ form $D^-$, the $u\bar{d}$ form $\pi^+$.

## (c) Kinematic rejection

Equal-and-opposite momenta $\Rightarrow$ lab is the mother rest frame. With $|\boldsymbol{p}| = 2.3182\,\mathrm{GeV}/c$, use $E^2 = p^2c^2 + m^2c^4$:

$$E_\pi \;=\; \sqrt{p^2c^2 + m_\pi^2c^4} \;=\; \sqrt{2.3182^2 + 0.13957^2}\,\mathrm{GeV}.$$

Evaluate:

$$E_\pi \;=\; \sqrt{5.3740 + 0.01948}\,\mathrm{GeV} \;=\; \sqrt{5.3935}\,\mathrm{GeV} \;=\; 2.3224\,\mathrm{GeV}.$$

Similarly for the $D^-$ ($m_D = 1.86966\,\mathrm{GeV}/c^2$):

$$E_D \;=\; \sqrt{2.3182^2 + 1.86966^2}\,\mathrm{GeV}.$$

Evaluate:

$$E_D \;=\; \sqrt{5.3740 + 3.4956}\,\mathrm{GeV} \;=\; \sqrt{8.8696}\,\mathrm{GeV} \;=\; 2.9782\,\mathrm{GeV}.$$

Sum (mother at rest):

$$M_{\mathrm{mother}}c^2 \;=\; E_\pi + E_D \;=\; 2.3224 + 2.9782\,\mathrm{GeV}.$$

$$M_{\mathrm{mother}}c^2 \;=\; \boxed{5.3006\,\mathrm{GeV}}. \tag{21}$$

Comparing to candidates ($B^0 = 5.277$, $B^{*0} = 5.325$, $B_1^0 = 5.721\,\mathrm{GeV}/c^2$) with $\sim 2\% \Rightarrow 0.1\,\mathrm{GeV}$ uncertainty: $B_1^0$ ruled out.

$$\boxed{B^0 \text{ and } B^{*0}.} \tag{22}$$

## (d) Line-shape selection

Plotted FWHM $\sim 1\,\text{milli-eV}/c^2$. Use $\Gamma = \hbar/\tau$ with $\hbar = 6.58\times10^{-16}\,\mathrm{eV\,s}$.

For $B^0$ with $\tau = 1.52\times10^{-12}\,\mathrm{s}$:

$$\Gamma_{B^0} \;=\; \frac{\hbar}{\tau_{B^0}} \;=\; \frac{6.58\times10^{-16}}{1.52\times10^{-12}}\,\mathrm{eV}.$$

Evaluate:

$$\Gamma_{B^0} \;=\; 4.3\times10^{-4}\,\mathrm{eV} \;=\; 0.43\,\text{milli-eV}. \tag{23}$$

For $B^{*0}$ with $\tau \sim 2\times10^{-15}\,\mathrm{s}$:

$$\Gamma_{B^{*0}} \;=\; \frac{6.58\times10^{-16}}{2\times10^{-15}}\,\mathrm{eV}.$$

Evaluate:

$$\Gamma_{B^{*0}} \;=\; 0.33\,\mathrm{eV} \;=\; 330\,\text{milli-eV}. \tag{24}$$

The plotted width matches $\Gamma_{B^0}$ and is $\sim 800\times$ too small for $B^{*0}$.

$$\boxed{\text{mother} = B^0.} \tag{25}$$

## (e) Sketch for $B^{*0}$

Lorentzian centred on $M_{\mathrm{inv}} - M_{B^{*0}} = 0$ with FWHM $\Gamma_{B^{*0}} \simeq 0.33\,\mathrm{eV}/c^2$, i.e. $\sim 800\times$ broader than the $B^0$ peak; vertical axis again normalised probability/arb.

### (f) Quark-model $(I, J^P)$

$d\bar{b}$: $b$ is isosinglet, $d$ has $I = \frac{1}{2}$, so $I = \frac{1}{2}$. $L = 0$ (lightest); fermion–antifermion parity $P = (-1)^{L+1} = -1$. Spin combinations $S = 0, 1 \Rightarrow J = 0, 1$. Standard pseudoscalar/vector identification:

$$\boxed{B^0 : (I, J^P) = (\tfrac{1}{2}, 0^-); \qquad B^{*0} : (I, J^P) = (\tfrac{1}{2}, 1^-).} \tag{26}$$

### (g) The decay $B^* \to B^0 + X$

Same flavour, same $P = -1$, $\Delta J = 1$: M1 radiative transition, $X = \gamma$. Two-body kinematics in the $B^*$ rest frame ($M_\gamma = 0$):

$$E_\gamma = \frac{M_{B^*}^2 - M_{B^0}^2}{2M_{B^*}} c^2.$$

Factor as a difference of squares:

$$E_\gamma = \frac{(M_{B^*} - M_{B^0})(M_{B^*} + M_{B^0})}{2M_{B^*}} c^2.$$

Substitute $M_{B^*} - M_{B^0} = 48.05\,\mathrm{MeV}/c^2$ and $M_{B^*} + M_{B^0} = 10601.37\,\mathrm{MeV}/c^2$, $2M_{B^*} = 10649.4\,\mathrm{MeV}/c^2$:

$$E_\gamma = \frac{48.05 \times 10601.37}{10649.4}\,\mathrm{MeV}.$$

Evaluate the numerator and divide:

$$E_\gamma = \frac{5.094 \times 10^5}{1.0649 \times 10^4}\,\mathrm{MeV}.$$

$$\boxed{E_\gamma \simeq 47.8\,\mathrm{MeV}.} \tag{27}$$

Slightly less than the splitting 48.05 MeV due to $B^0$ recoil.

### (h) Collider $\sqrt{s}$

Symmetric collider $\Rightarrow$ each $B$ carries $\sqrt{s}/2$, so

$$\gamma = \frac{\sqrt{s}}{2Mc^2}, \qquad \beta\gamma = \sqrt{\gamma^2 - 1}.$$

Mean lab decay length:

$$L = \beta\gamma c\tau.$$

With $\tau_{B^0} = 1.52\,\mathrm{ps}$, $c\tau = (3 \times 10^8)(1.52 \times 10^{-12})\,\mathrm{m} = 4.56 \times 10^{-4}\,\mathrm{m} = 0.456\,\mathrm{mm}$. Solve for $\beta\gamma$ using $L_0 = 0.2\,\mathrm{mm}$:

$$\beta\gamma = \frac{L_0}{c\tau} = \frac{0.2}{0.456} = 0.439.$$

Square and add 1:

$$\gamma^2 = 1 + (\beta\gamma)^2 = 1 + 0.193 = 1.193.$$

Take the square root:

$$\gamma = \sqrt{1.193} = 1.092.$$

Then

$$\sqrt{s} = 2\gamma Mc^2 = 2 \times 1.092 \times 5.27666\,\mathrm{GeV}.$$

$$\sqrt{s} \simeq 11.5\,\mathrm{GeV}, \tag{28}$$

$$\boxed{\sqrt{s} \simeq 10.6\text{–}11.5\,\mathrm{GeV}} \tag{29}$$

i.e. the $\Upsilon(4S)$ at 10.58 GeV (BaBar/Belle/Belle II $B$-factories).